

CO4 - AT-I10) Unified big data architecture for retail:-

- * A retail organisation can use a data lake architecture to combine data from websites, mobile apps, physical stores.
- * This creates a 360 degree customer view and support personalised recommendations and targeted marketing.

proposed architecture:-

- Data sources.
- Data ingestion.
- Processing
- Analytics and ML//.

20) Distributed computing for specific research data

- * Scientific research often produces very large datasets that cannot be efficiently processed by a single computer.
- * Parallel processing allows these computers to work simultaneously.

Advantages:-

- Faster processing.
- Better scalability.
- Handles large datasets//.

30) Fraud detection framework for digital payments:-

- * A fraud detection framework should process transaction data in real time.

Payment transaction → Streaming platform → Real-time processing →
ML fraud model → Fraud score → Alert / Block.

Technologies:-

- Apache Kafka
- Apache Spark Streaming
- Machine learning models.
- Database//.

40) Storing and processing multimedia data:-
* Multimedia data such as images, audio, video creates challenges because of its large size, high processing requirements & different formats.

Challenges:-

- * High storage requirement.
- * Slow data transfer.
- * Different file formats.
- * Metadata management.

Recommended technologies:-

- * Cloud object storage.
- * HDFS.
- * Apache Spark.

50) Disaster recovery and Business Strategy:-

* A cloud-based big data platform should maintain continuous operations even when failures occur.

Strategy:-

- 1) Store data in multiple.
- 2) Maintain primary.
- 3) Secondary systems.
- 4) Use automated backups.

Key machines:-

- Redundancy.
- Replication.
- Automated failure.

60) Data quality and data governance:-

* Poor data quality reduces the reliability of analytical results.

Major issues:-

Inconsistency, Duplication, Missing values.

Data governance strategies:-

- Data verification rule.
- Data standardization.
- Missing-value handling.

Scalable recommendation system for streaming services

* A streaming platform can analyze user behaviours to recommend relevant content.

User activity - Kafka → Datalake → Spark processing → recommendation model → personalized recommendations.

Data analyze:-

- Viewing history.
- Watch time.
- Likes and ratings.
- Content preferences.

80) Role of cloud computing in big data

cloud computing provides infrastructure and services for storing and processing large volumes of data.

Advantages:-

- Scalability.
- Flexibility.
- Cost efficiency.
- Global access.

90) Real-time analytics for transportation and logistics

* A transportation company can collect real-time data from GPS devices, vehicles, sensors, traffic systems and fuel monitoring systems.

GPS/sensors → Kafka → Stream processing → analytics → route optimization system → Dashboard.

100) Ethical and legal challenges in big data

* Big data collection creates important ethical & legal concerns.

Challenges:-

- * Privacy violations.
- * Data breaches.

* Lack of transparency//.

Recommended policies:-

* Collect only necessary data.

* Obtain clear user consent.

* Use encryption & access control.

* Perform regular security audits//.